

Non interacting System

- assume const density
- const cross sectional area
- linear flow resistance

Find transfer function relating h_2 to q $\frac{H_2(s)}{Q(s)}$

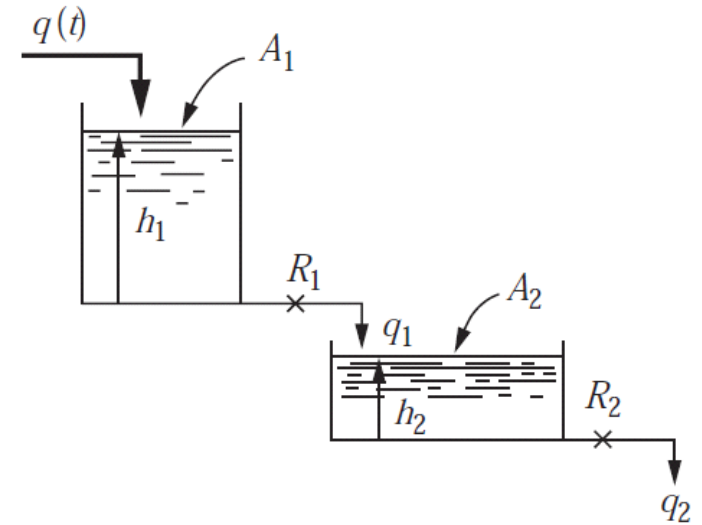
Found transfer func for each tank
and cancel out intermediate flow

Balance for tank 1

$$q - q_1 = A_1 \frac{dh_1}{dt}$$

tank 2

$$q_1 - q_2 = A_2 \frac{dh_2}{dt}$$



with $q_1 = \frac{h_1}{R_1}$; $q_2 = \frac{h_2}{R_2}$

using $\frac{H(s)}{Q(s)} = \frac{R}{s+1}$ and $\frac{Q_o(s)}{Q(s)} = \frac{1}{s+1}$

$$\Rightarrow \frac{H_2(s)}{Q_1(s)} = \frac{R_2}{s+1} \quad \frac{Q_2(s)}{Q_1(s)} = \frac{1}{s+1}$$

Multiplying the two

$$\frac{H_2(s)}{Q_1(s)} \frac{Q_1(s)}{Q(s)} = \boxed{\frac{R_2}{\tau_2 s + 1} \frac{1}{\tau_1 s + 1}} = \frac{H_2(s)}{Q(s)}$$

For unit change in input $Q(s) = \frac{1}{s}$

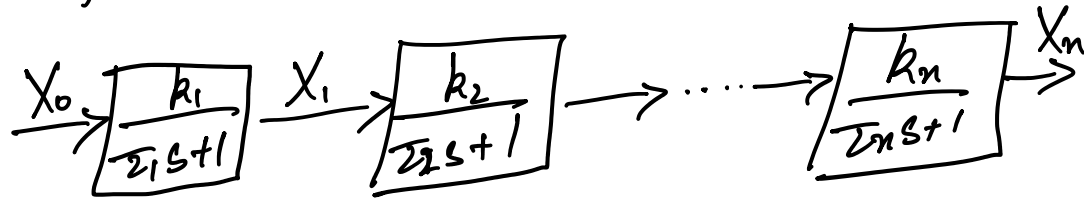
$$H_2(s) = \frac{1}{s} \frac{R_2}{(\tau_1 s + 1)(\tau_2 s + 1)}$$

Inversion using partial fractions

$$H_2(t) = R_2 \left[1 - \frac{\tau_1 \tau_2}{\tau_1 - \tau_2} \left(\frac{1}{\tau_2} e^{-t/\tau_1} - \frac{1}{\tau_1} e^{-t/\tau_2} \right) \right]$$

Generalization for Several Non interacting Systems in Series

for n non interacting first order sys



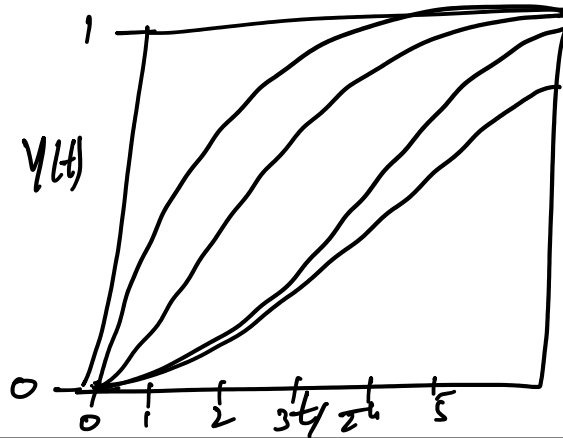
$$\frac{X_1(s)}{X_0(s)} = \frac{k_1}{\tau_1 s + 1}$$

$$\frac{X_2(s)}{X_1(s)} = \frac{k_2}{\tau_2 s + 1}$$

$\vdots$

$$\frac{X_n(s)}{X_{n-1}(s)} = \frac{k_n}{\tau_n s + 1}$$

$$\frac{X_n(s)}{X_0(s)} = \prod_{i=1}^n \frac{k_i}{\tau_i s + 1}$$



Interacting System

Mass balance over both tanks

Tank 1 $q - q_1 = A_1 \frac{dh}{dt} \quad @ \text{S.S. } q_s - q_{1s} = 0$

Tank 2 $q_1 - q_2 = A_2 \frac{dh}{dt} \quad @ \text{S.S. } q_{1s} - q_{2s} = 0$

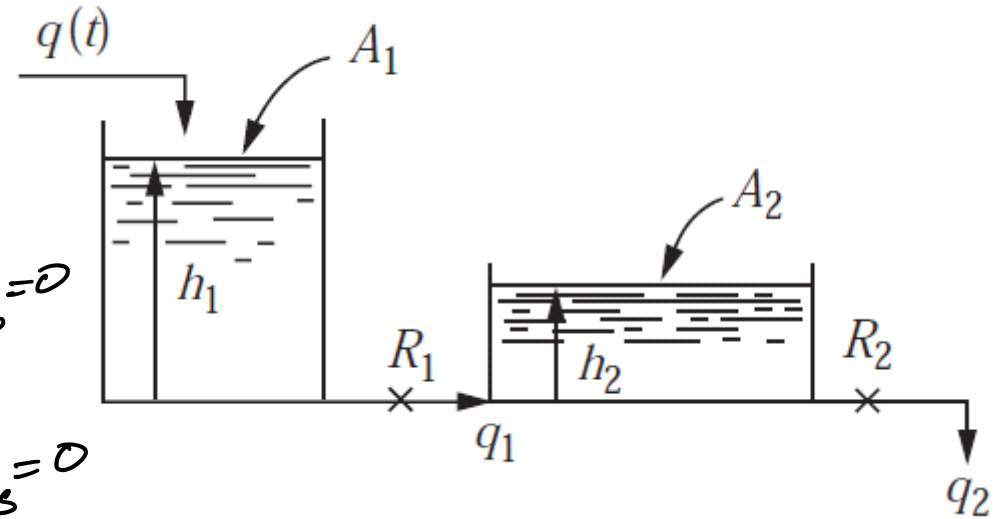
$\Rightarrow$ Tank 1 $Q - Q_1 = A_1 \frac{dH_1}{dt}$

Tank 2 $Q_1 - Q_2 = A_2 \frac{dH_2}{dt}$

Valve 1

$R_1 = \frac{h_1 - h_2}{q_1} \Rightarrow Q_1 = \frac{H_1 - H_2}{R_1}$

Valve 2 $R_2 = \frac{h_2}{q_2} \Rightarrow Q_2 = \frac{H_2}{R_2}$



Laplace of all

Tank 1 $Q(s) - Q_1(s) = A_1 s H_1(s)$

Tank 2 $Q_1(s) - Q_2(s) = A_2 s H_2(s)$

Valve 1 $Q_1(s) = \frac{H_1(s) - H_2(s)}{R_1}$

Valve 2 $Q_2(s) = \frac{H_2(s)}{R_2}$

$$\Rightarrow H_1(s) = \frac{Q(s) - Q_1(s)}{A_1 s}$$

$$\Rightarrow \frac{H_1(s) - H_2(s)}{R_1} - \frac{H_2(s)}{R_2} = A_2 s H_2(s)$$

$$H_1(s) = \frac{Q(s) - \left(\frac{H_1(s) - H_2(s)}{R_1} \right)}{A_1 s}$$

$$A_1 s H_1(s) = Q(s) - \left(\frac{H_1(s) - H_2(s)}{R_1} \right)$$

$$H_1(s) \left(A_1 s + \frac{1}{R_1} \right) = Q(s) + \frac{H_2(s)}{R_1}$$

$$H_1(s) = \frac{Q(s) + \frac{H_2(s)}{R_1}}{A_1 s + \frac{1}{R_1}}$$

Placing in to second eqn

$$\frac{Q(s) + \frac{H_2(s)}{R_1}}{A_1 s + \frac{1}{R_1}} - H_2(s) - \frac{H_2(s)}{R_2} = A_2 s H_2(s)$$

$$\frac{1}{R_1} \left[\frac{Q(s)}{A_1 s + \frac{1}{R_1}} + \frac{H_2(s)}{R_1 (A_1 s + \frac{1}{R_1})} - H_2(s) \right] - \frac{H_2(s)}{R_2} = A_2 s H_2(s)$$

$$\frac{Q(s)}{R_1 (A_1 s + \frac{1}{R_1})} = H_2(s) \left[A_2 s + \frac{1}{R_2} + \frac{1}{R_1} - \frac{1}{R_1^2 (A_1 s + \frac{1}{R_1})} \right]$$

$$\frac{H_2(s)}{Q(s)} = \frac{1}{R_1 (A_1 s + \frac{1}{R_1})} \times \frac{1}{A_2 s + \frac{1}{R_1} + \frac{1}{R_2} - \frac{1}{R_1^2 (A_1 s + \frac{1}{R_1})}}$$

$$= \frac{1}{R_1 \left(\frac{A_1 R_1 s + 1}{R_1} \right)} \times \frac{1}{A_2 s + \frac{1}{R_1} + \frac{1}{R_2} - \frac{1}{R_1^2 \left(\frac{s+1}{R_1} \right)}}$$

$$= \frac{1}{s+1} \times \frac{1}{A_2 s + \frac{1}{R_1} + \frac{1}{R_2} - \frac{1}{R_1 (s+1)}} = \frac{1}{s+1} \times \frac{1}{\frac{R_1 R_2 A_2 s + R_2 + R_1}{R_1 R_2} - \frac{1}{R_1 (s+1)}}$$

$$= \frac{1}{s+1} \times \frac{1}{\frac{R_1 R_2 A_2 s + R_1 + R_2}{R_1 R_2} - \frac{1}{R_1 (s+1)}} = \frac{1}{\cancel{s+1}} \times \frac{R_1 R_2 (s+1)}{(s+1) (R_1 R_2 A_2 s + R_1 + R_2) - R_2}$$

$$\frac{H_2(s)}{Q(s)} = \frac{R_1 R_2}{(\tau_1 s + 1)(R_1 \tau_2 s + R_1 + R_2) - R_2}$$

$$\begin{aligned} & \tau_1 \tau_2 R_1 s^2 + \tau_1 s R_1 + \tau_1 s R_2 + R_1 \tau_2 s + R_1 + R_2 - R_2 \\ &= (\tau_1 \tau_2 R_1 s^2 + s(\tau_1 R_1 + \tau_1 R_2 + R_1 \tau_2) + R_1) \frac{R_1}{R_1} \\ &= \tau_1 \tau_2 s^2 + s(\tau_1 + \frac{\tau_1 R_2}{R_1} + \tau_2) + 1 \\ &= \tau_1 \tau_2 s^2 + s(\tau_1 + A_1 R_2 + \tau_2) + 1 \end{aligned}$$

$$\frac{H_2(s)}{Q(s)} = \frac{R_2}{\tau_1 \tau_2 s^2 + (\tau_1 + \tau_2 + A_1 R_2) s + 1}$$

Different from multiplying transfer functions
 Interacting $\rightarrow$ Loading
 Second tank loads the first tank

For non interacting system

$$\frac{Q_2(s)}{Q(s)} = \left(\frac{1}{\tau_1 s + 1} \right) \left(\frac{1}{\tau_2 s + 1} \right)$$

if both have same τ

$$\frac{Q_2(s)}{Q(s)} = \left(\frac{1}{\tau s + 1} \right)^2$$

if change is unit step $Q(s) = \frac{1}{s}$

$$Q_2(s) = \frac{1}{s} \left(\frac{1}{\tau s + 1} \right)^2 = \frac{A}{s} + \frac{B}{\tau s + 1} + \frac{C}{(\tau s + 1)^2}$$

$$\Rightarrow 1 = A(\tau s + 1)^2 + Bs(\tau s + 1) + Cs$$

$$1 = A\tau^2 s^2 + A2\tau s + A + B\tau s^2 + Bs + Cs$$

$$1 = s^2(A\tau^2 + B\tau) + s(A2\tau + B + C) + A$$

$$\hookrightarrow A\tau^2 + B\tau = 0, \quad A2\tau + B + C = 0, \quad A = 1$$

$$\tau^2 + B\tau = 0$$

$$B = -\frac{\tau^2}{\tau} = -\tau$$

$$2\tau - \tau + C = 0$$

$$C = \tau - 2\tau$$

$$C = -\tau$$

Putting back in eqn

$$Q_2(s) = \frac{1}{s} + \frac{-\tau}{\tau s + 1} + \frac{-\tau}{(\tau s + 1)^2} \rightarrow \frac{\tau^2 (s + \frac{1}{\tau})^2}{\tau^2 (s + \frac{1}{\tau})^2}$$

$$\text{using } \mathcal{L}^{-1}\left\{\frac{1}{(s+a)^2}\right\} = t e^{-at}$$

$$Q_2(s) = \frac{1}{s} - \frac{1}{s + \frac{1}{\tau}} - \frac{1}{\tau (s + \frac{1}{\tau})^2}$$

$$Q_2(t) = 1 - e^{-t/\tau} - \frac{t}{\tau} e^{-t/\tau}$$

When system is interacting
with $A_1=A_2$ and $\tau_1=\tau_2=\tau$

$$\frac{Q_2(s)}{Q(s)} = ?$$

with as^2+bs+c
and $s = \frac{-b \pm \sqrt{b^2-4ac}}{2a} = \frac{-3\tau \pm \sqrt{(3\tau)^2 - 4(\tau^2)(1)}}{2(\tau^2)}$
 $= \frac{-3\tau \pm \sqrt{5\tau^2}}{2\tau^2} = \frac{-3 \pm \sqrt{5}}{2\tau}$

with

$$\frac{H_2(s)}{Q(s)} = \frac{R_2}{\tau_1\tau_2 s^2 + (\tau_1 + \tau_2 + A_1 R_2)s + 1}$$

and $R_2 Q_2(s) = H_2(s)$

$$\frac{Q_2(s)}{Q(s)} = \frac{1}{\tau_1\tau_2 s^2 + (\tau_1 + \tau_2 + A_1 R_2)s + 1}$$

with $\tau_1 = \tau_2 = \tau$

$$\frac{Q_2(s)}{Q(s)} = \frac{1}{\tau^2 s^2 + 3\tau s + 1} \Rightarrow \frac{1}{(0.38\tau s + 1)(2.62\tau s + 1)}$$

$$\Rightarrow \tau^2 s^2 + 0.381\tau s + 2.62\tau s + 1$$

$$= \tau^2 s^2 + 2.62\tau s + 0.381\tau s + 1$$

$$= 2.62\tau s(0.381\tau s + 1) + 1(0.381\tau s + 1)$$

$$= (0.38\tau s + 1)(2.62\tau s + 1)$$

$\frac{3-\sqrt{5}}{2} = 0.381$
 $\frac{3+\sqrt{5}}{2} = 2.61$

For unit step change

$$Q_2(s) = \frac{1}{s} \frac{1}{(0.38\tau s + 1)(2.62\tau s + 1)} = \frac{A}{s} + \frac{B}{0.38\tau s + 1} + \frac{C}{2.62\tau s + 1}$$

$$\Rightarrow 1 = A(0.38\tau s + 1)(2.62\tau s + 1) + B s(2.62\tau s + 1) + C s(0.38\tau s + 1)$$

$$1 = A(0.996\tau^2 s^2 + 0.38\tau s + 2.62\tau s + 1) + B 2.62\tau s^2 + B s + C 0.38\tau s^2 + C s$$

$$1 = A 0.996\tau^2 s^2 + A 3\tau s + A + B 2.62\tau s^2 + B s + C 0.38\tau s^2 + C s$$

$$1 = s^2(A 0.996\tau^2 + B 2.62\tau + C 0.38\tau) + s(3A\tau + B + C) + A$$

$$\Rightarrow A = 1 ; \quad 3A\tau + B + C = 0 \quad ; \quad A 0.996\tau^2 + B 2.62\tau + C 0.38\tau = 0$$

$$3\tau + B + C = 0$$

$$B = -C - 3\tau$$

$$B = 0.0625\tau$$

$$0.996\tau^2 - C 2.62\tau - 7.86\tau^2 + C 0.38\tau = 0$$

$$-6.86\tau^2 - 2.24\tau C = 0$$

$$C = \frac{6.86\tau^2}{-2.24\tau} = -3.0625\tau$$

$$= \frac{1}{s} + \frac{0.0645\tau}{0.38\tau s + 1} - \frac{3.0645\tau}{2.62\tau s + 1}$$

$$= \frac{1}{s} + \frac{0.17 \times 0.38\tau}{0.38\tau s + 1} - \frac{1.17 \times 2.62\tau}{2.62\tau s + 1}$$

$$= \frac{1}{s} + 0.17 \frac{1}{s + \frac{1}{0.38\tau}} - 1.17 \frac{1}{s + \frac{1}{2.62\tau}}$$

$$Q_2(t) = 1 + 0.17 e^{-\frac{t}{0.38\tau}} - 1.17 e^{-\frac{t}{2.62\tau}}$$

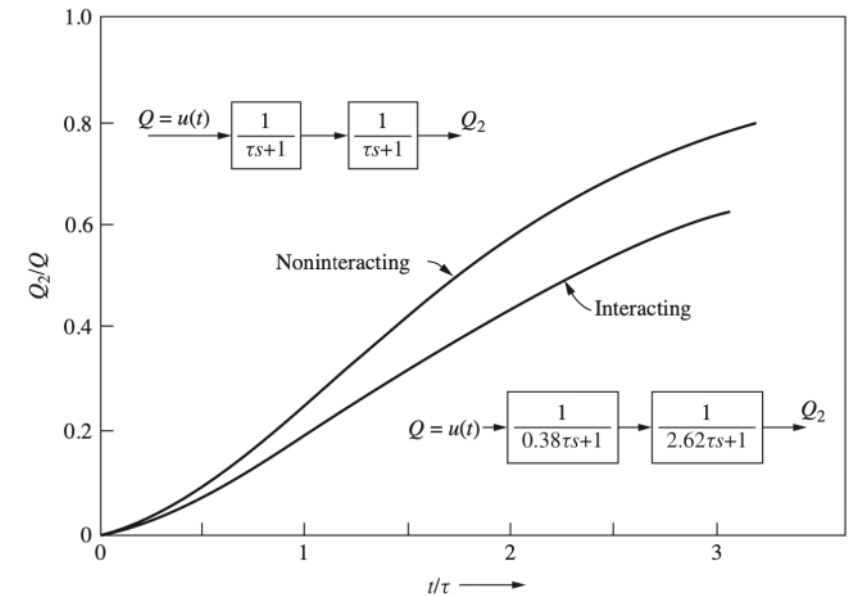


FIGURE 6-7
Effect of interaction on step response of two-tank system.